Everyday spreadsheet work

First wireframe set: familiar desktop editing, visible outcomes, and a path from entry to export.

18 screens and review sheets

A native desktop layout built around standard Qt Widgets. These are static design proposals, not screenshots of an implemented application.

01 Start in the grid

A blank sheet is ready for typing. SQL and Python remain optional tools.

02 Keep outcomes visible

Saved data, pending calculations, rollback, and failed follow-up work have distinct messages.

03 Use desktop conventions

Menus, compact toolbars, keyboard focus, resizable panels, and task dialogs.

Review at fit-to-width; the PDF contains selectable text and vector linework.

READING MAP

02	Main workbook window
03-05	Create, navigate, edit, copy, and fill
06-08	Calculate, format, and organize
09-11	Sort, filter, find, and undo
12-15	Import, save, workbook rules, and export
16	Walk through the Supplies session
17	Qt Widgets fit and platform behavior
18	Story coverage and review decisions

Status: Draft 0.1. Based on DH-US-01 through DH-US-20. The fixture uses Notebook, Pen, and Folder; its total starts at 18.00.

02 / Main workbook window

The primary workspace stays centered on the sheet. Optional tools live in panels.

DH-US-01, 04-06, 09, 12-14

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | 1 | Sort / Filter

D6 | fx | =SUM(D2:D4) 2

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		
5						
6	**Total**			18.00		
7						
8						
9						
10						
11						
12						

3

< > | Supplies | Notes | + | 4

Saved | Calculations current | 5 | 100%

1 One action model

Menus and compact toolbars expose the same commands. Advanced tools open from Tools; they do not occupy the default view.

2 Address and formula

The address box accepts a cell or range. The formula bar always exposes source, while the grid shows the result.

3 Grid first

Row and column headers, active-cell outline, selection fill, a fill handle, and ordinary scrollbars.

4 Sheet navigation

Tabs support add, rename, reorder, duplicate, and delete. A tab list handles overflow.

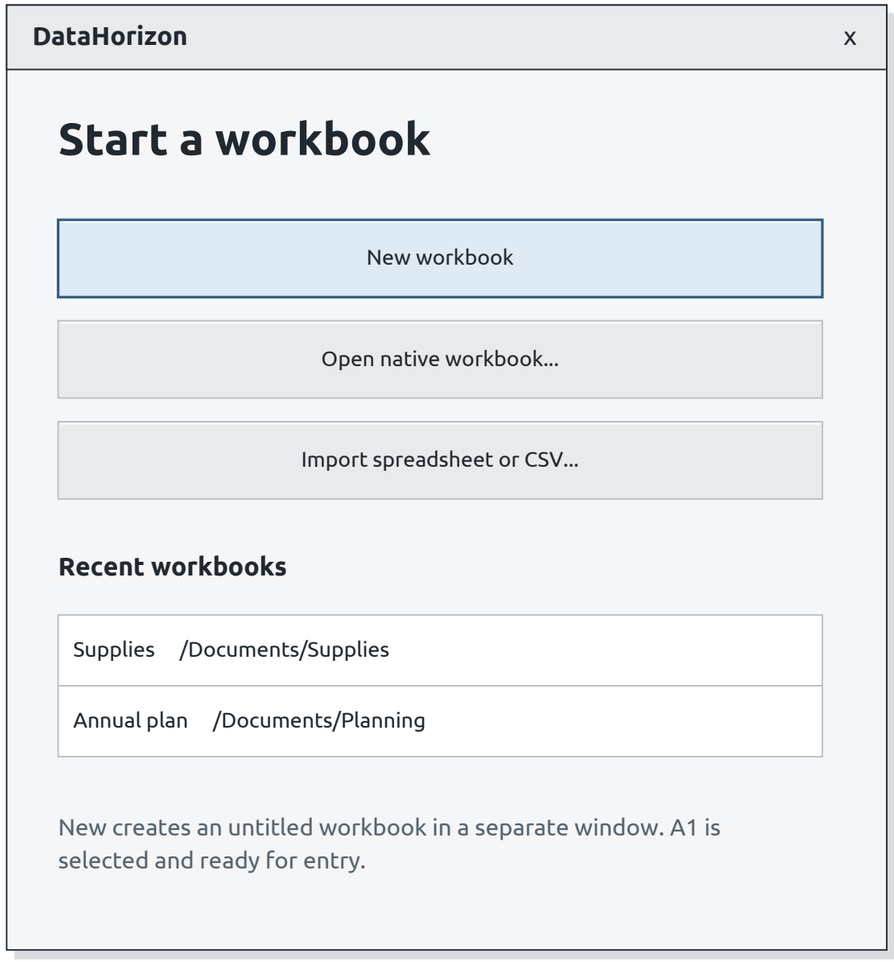
5 Two independent states

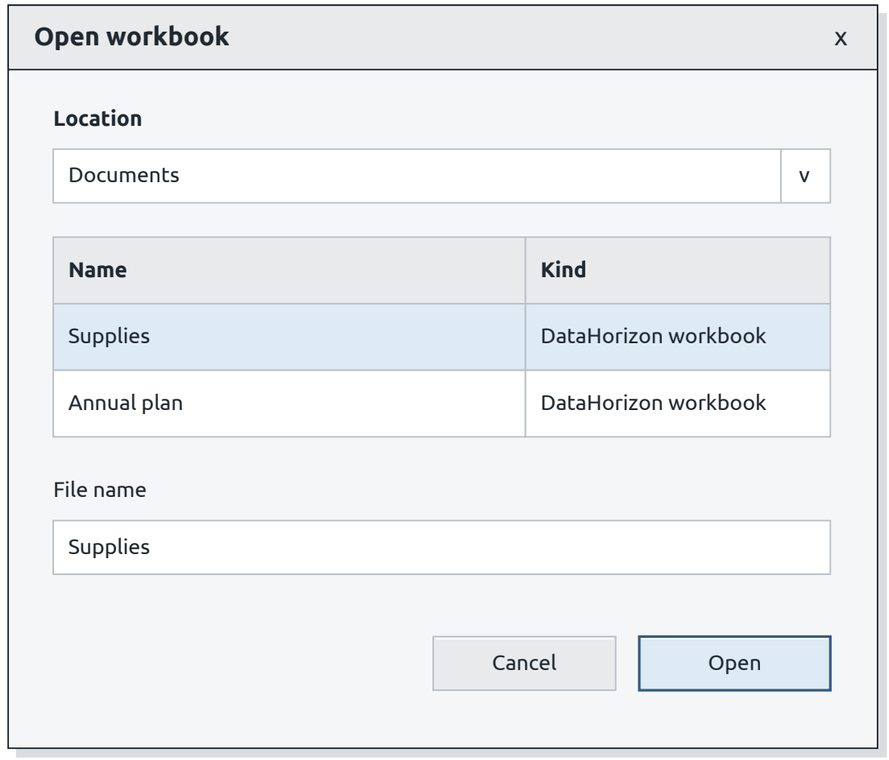
Saved describes stored inputs and presentation. Calculations current describes derived results.

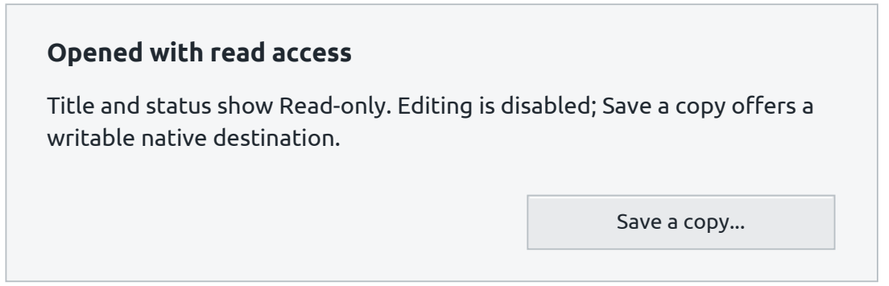
03 / Create, open, and continue

Start without schema setup. Use native file dialogs for local workbook selection.

DH-US-01, 02, 04







1 Blank means editable

The first key types into A1. Initial naming and native location can wait until Save.

2 Native file selection

The file list is schematic. The operating system supplies its actual dialog, file type labels, and navigation.

3 Safe open errors

Missing, incompatible, or unreadable files show the reason and actions to choose another file. Opening never overwrites the source.

4 Reopen freshness

Saved formula results remain unverified until refreshed. Show Calculating or an explicit error; never label them current by default.

04 / Select, enter, and edit

The same source appears in the cell editor and formula bar. Escape cancels the current draft.

DH-US-05, 06

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

B3 **fx** 6

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	6	1.25	5.00		
4	Folder	3	2.00	6.00		
5						
6	**Total**			18.00		
7						
8						

< > Supplies | Notes | +

Editing B3 Previous stored result shown 100%

1 Edit modes

Typing replaces the selected value. F2 or double-click edits existing source. Enter commits; Escape restores the pre-edit content.

2 Navigation

Tab commits and moves right; Enter moves down. Focus returns to the grid after an accepted edit.

3 Value intent

Text format preserves 00123. Number and date conversion follows the declared locale and compatibility rules.

4 Selection without edits

Address jumps and viewport reads do not run change handlers. Click headers for whole rows or columns; corner selects the sheet.

Jump to a range

D5000:F5010 | Go to range

Arrow keys move; Shift extends a selection.

Clear contents

Delete clears values and formulas in the selected range. Format > Clear formatting is a separate command.

A text number: enter 00123 using Text format.

05 / Copy, paste, and fill

Show the source and destination. Ordinary range operations succeed or roll back as a unit.

DH-US-07, 08, 18

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

D2:D4 **fx** =B2*C2

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		
5						
6	**Total**			18.00		

< > Supplies | Notes | +

Ready to fill Calculations current 100%

Fill selection x

Destination: Supplies!D2:D4

(•) Copy first formula / value

( ) Continue numeric series

Cancel | Fill

Paste special x

Destination: Supplies!F2:G4

(•) Values only

( ) Formats only

Cancel | Paste

1 Two ways to fill

Drag the handle or use Edit > Fill. The destination is highlighted before release or confirmation.

2 References

D2 =B2*C2 becomes D3 =B3*C3. An absolute component, such as B1, stays fixed. Exact reference rules remain to be specified.

3 Clipboard behavior

Normal paste keeps supported formulas and formats. Tab-separated text maps to cells; Paste special is explicit.

4 Recoverable operations

Cut removes the source only with a successful destination write. A rejected paste or fill changes neither range. Each accepted operation is one undo item.

06 / Formulas, pending results, and errors

A result has an explicit freshness state. An error keeps its formula available for correction.

DH-US-09, 10

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

D6 | fx | =SUUM(D2:D4)

Formula error at D6: Unknown function SUUM. Inspect the formula below.

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	6	1.25	7.50		
4	Folder	3	2.00	6.00		
5						
6	**Total**			#NAME?		

< > | Supplies | Notes | +

Saved 1 formula error 100%

Issues - Formula errors x

Supplies!D6 Unknown function: SUUM

`Formula: =SUUM(D2:D4)`

Use a supported function, or keep editing.

Edit D6

1 Source survives

The formula bar retains =SUUM(D2:D4). The cell displays an error, never a fabricated zero.

2 Pending is visible

After an input edit, dependent cells show Calculating... instead of an unmarked old number. Status reads Saved | Calculating.

3 Correct and recover

Changing SUUM to SUM recalculates D6 to 20.50. Circular references show involved cells in the same Issues panel.

4 Freshness contract

A result from an older input state cannot replace a newer result. Function support is staged; shown functions are proposed first-tier examples.

07 / Formatting and large-sheet readability

Separate stored meaning from display. View changes remain available through ordinary menus.

DH-US-13, 14

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

C2:D4 **fx** 3.5

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		

< > | Supplies | Notes | +

Saved Calculations current 100%

Format cells x

Number **Font** **Alignment** **Borders** **Fill**

Category: Number v Decimals: 2 v

Preview: 3.50 Stored value: 3.5

Cancel | OK

View options x

- [x] Freeze first row
- [ ] Freeze first column
- [ ] Wrap selected cells

Apply

1 Common formatting

Toolbar shortcuts cover emphasis, borders, fill, and common number formats. Format cells exposes the rest.

2 Meaning is preserved

3.5 displays as 3.50. Clearing formatting retains the number or formula. Formats-only paste changes presentation.

3 Frozen and hidden

A thicker divider marks a frozen boundary. Hidden column D shows a gap marker between C and E with an Unhide action.

4 Bounded views

Resize headers by dragging. Fit column uses loaded or explicitly sampled content. Distant navigation fetches a viewport, not the entire workbook.

08 / Sheets, rows, and columns

Structure commands stay close to the object they affect, with consequences visible before deletion.

DH-US-11, 12

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

A3:D3 **fx**

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		
5						
6	**Total**			18.00		

< > | Supplies | Notes | +

Saved Calculations current 100%

Supplies tab context menu

- Rename...
- Duplicate...
- Move left
- Move right
- Delete sheet...

Delete referenced sheet? x

Notes contains formulas that use Supplies.

Deleting it will leave broken references (#REF!).

Cancel | Delete sheet

1 Sheet tabs

Add with +. Rename by menu or double-click; drag to reorder. Duplicate gets a distinct name and supported content and formatting.

2 Rows and columns

Header context menus contain Insert before, Delete, Resize, Hide, and Unhide. Commands apply to the full selected structure.

3 Reference consequences

Insertion follows surviving cells; deleting a direct target yields a broken reference. Show affected references when known.

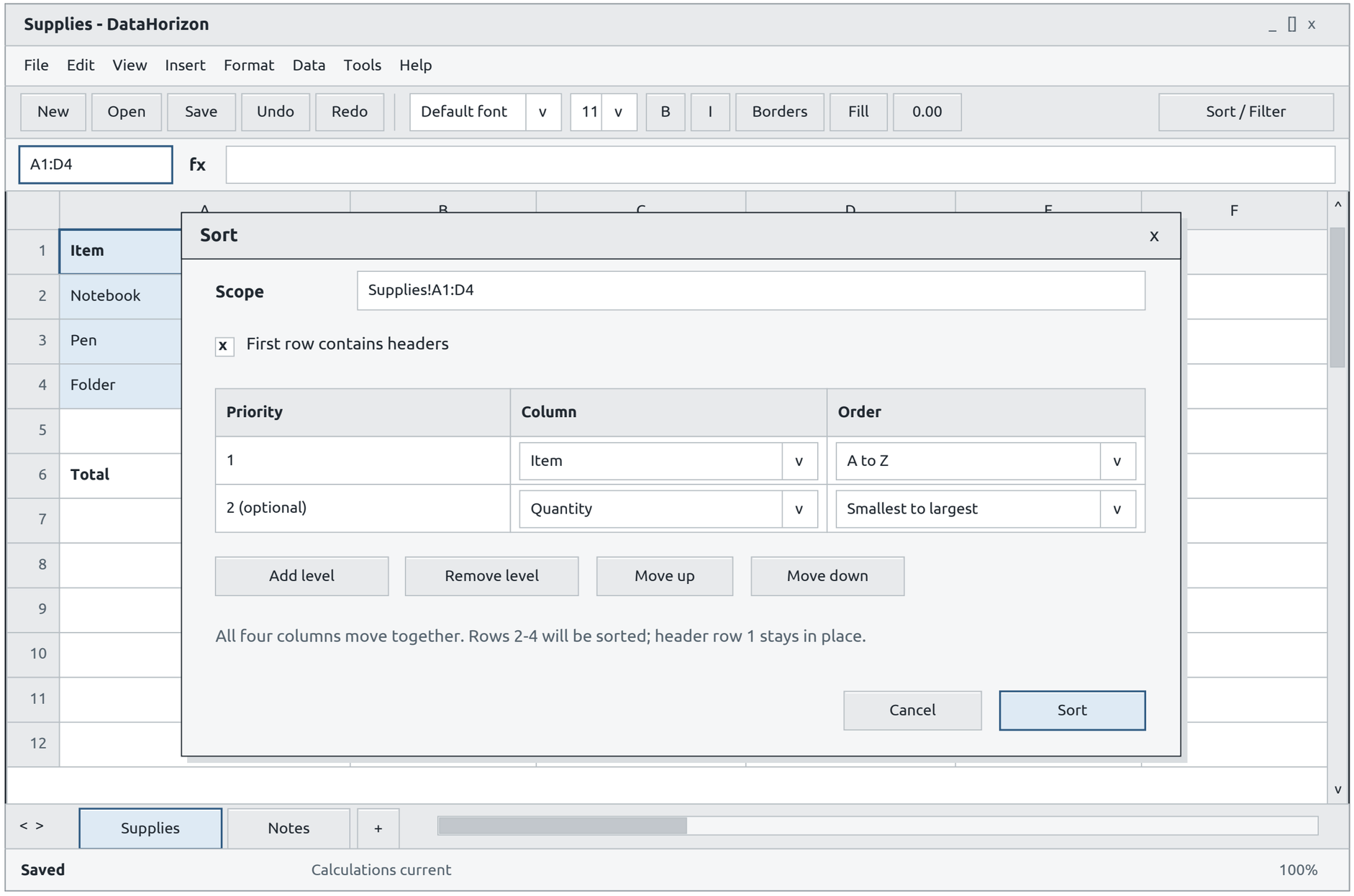
4 Rename versus duplicate

Rename preserves the sheet identity. Duplicate-sheet reference targets and exact sort/reference rules remain design decisions.

09 / Sort a declared list

Choose the whole record scope and header treatment before changing order.

DH-US-15



1 Declare the scope

Select a list or table. If only one column is selected beside related data, offer Expand selection before sorting.

2 Keep records intact

IDs, labels, quantities, prices, and formulas move as complete records within the declared list.

3 Multiple keys

Add or reorder priority levels. The effective key order is visible before applying.

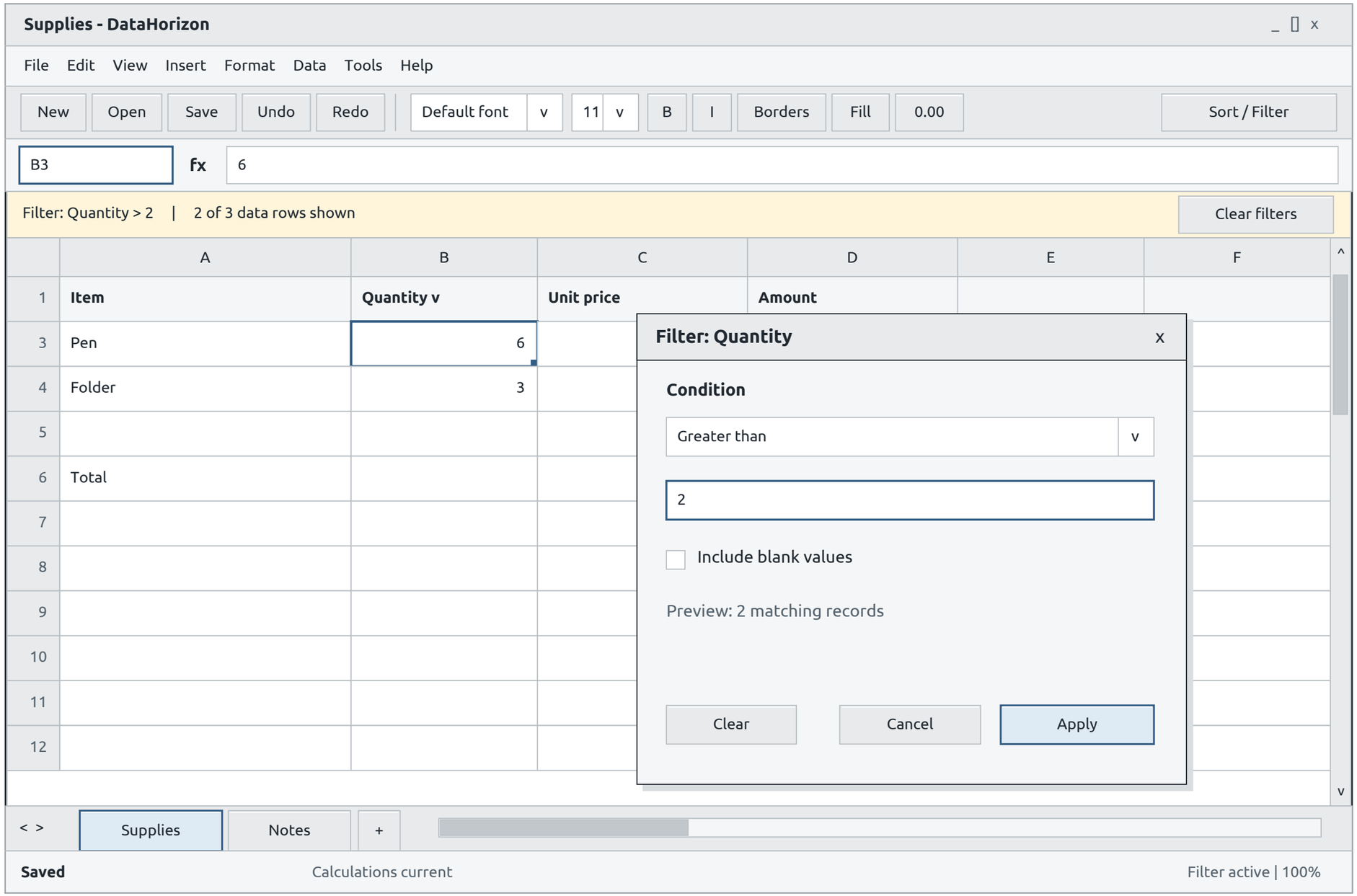
4 Reference rules to resolve

Stable record identity is required. Exact freeform formula movement and reference adjustment need an explicit contract before implementation.

10 / Filter without deleting

Filter criteria stay visible, and the user can always return to all rows.

DH-US-16



1 Visible criteria

Show the condition and matching / total count above the grid. Clear filters restores all stored rows.

2 Record targets

Filtered displays retain source row labels or stable record context. Editing targets the intended record.

3 An edit can hide a row

If Quantity changes from 6 to 1, accept the edit and say Row no longer matches filter. Offer Clear filters.

4 Empty is distinguishable

For no matches show No rows match Quantity > 2 with Clear filters. Do not imply the dataset has been deleted.

11 / Find, replace, and undo

A docked tool keeps search scope and location visible while working in the grid.

DH-US-17, 18

Supplies - DataHorizon _ ▯ x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | More...

A3 | fx | Pen

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		
5						
6	**Total**			18.00		
7						
8						
9						
10						
11						
12						

< > | Supplies | Notes | +

Saved Calculations current 100%

Find and replace x

Find

Pen

Replace with

Blue pen

Scope

Active sheet: Supplies v

Search in

Displayed values v

☐ Match case

☐ Whole cell only

1 match: Supplies!A3

Previous | Next

Replace | Review all...

1 matching cell on Supplies. Replacements will be checked by workbook rules.

1 Scope before change

Selection, active sheet, or workbook. Displayed values and formula source are distinct search modes.

2 Review Replace All

Show the final scope and match count before one managed Replace All operation. Validation rejection leaves all affected cells unchanged.

3 Named undo action

Edit > Undo Paste D2:D4 describes what will reverse. One paste or fill is one action; redo disappears after an intervening edit.

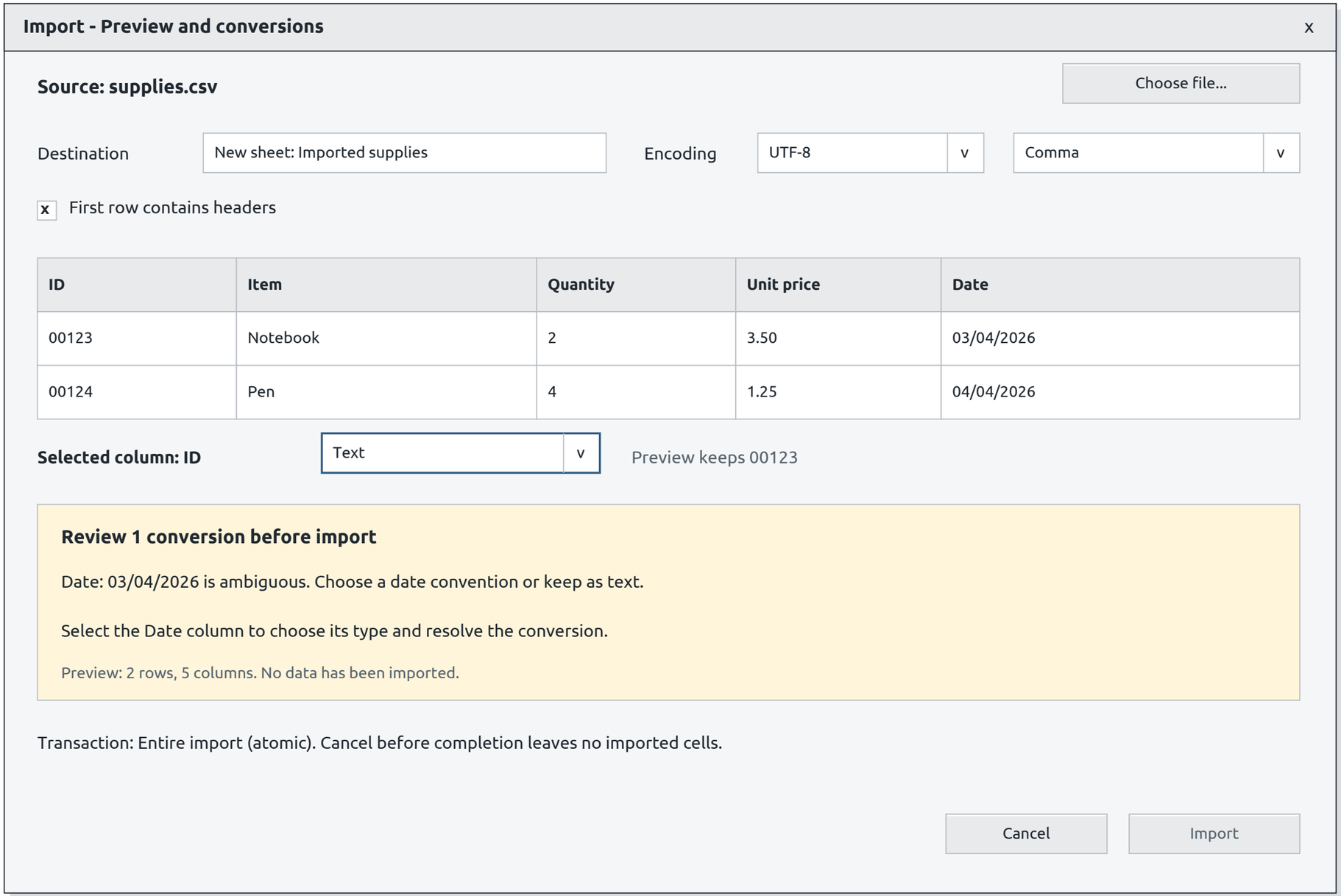
4 Limits are explicit

If undo is unavailable, explain why. Reversal runs applicable rules and may be rejected. External side effects are outside ordinary edit undo.

12 / Import with a conversion preview

Stage the source, choose types, and inspect losses before applying managed changes.

DH-US-03



1 Explicit types

Column type selection prevents lost leading zeros. Ambiguous dates and conversion failures show locations and examples.

2 Compatibility report

Inspect values, formulas, formatting, and other features separately. Resolve conversions before Import is enabled. Imported workbook code stays inactive.

3 Transaction scope

This screen shows an atomic import. A future chunked mode needs a distinct review stating committed batches and cancellation behavior.

4 Result and progress

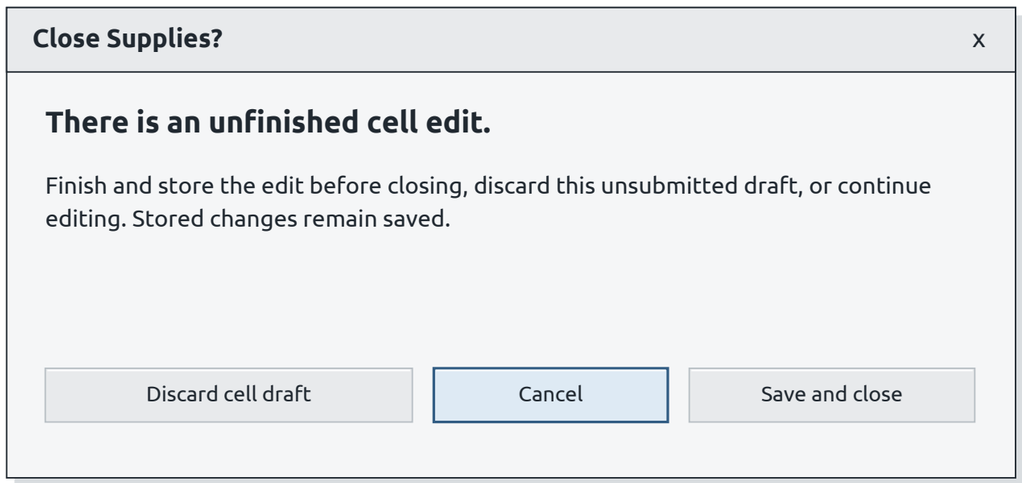
After Apply show imported cells and warnings. For a long atomic import, progress remains cancellable until commit; success reports the actual outcome.

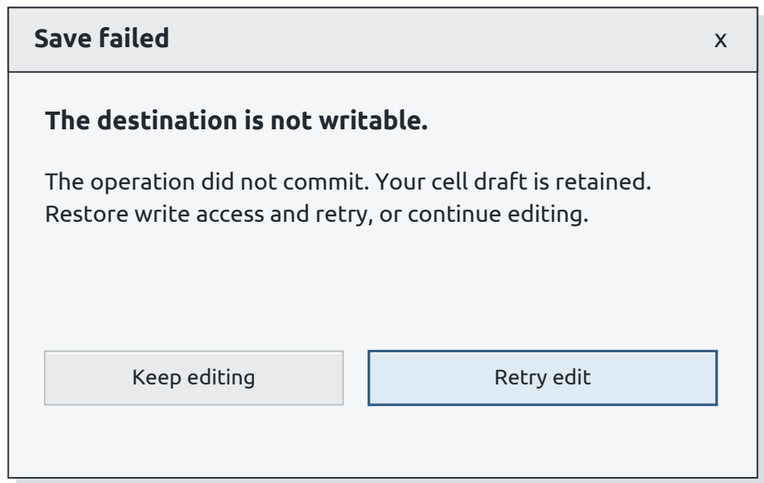
13 / Save, pending work, and close

Proposal: store each accepted operation automatically; Save finishes pending edits and confirms storage.

DH-US-02, 04, 10

Status shown	Meaning	Next action
Editing B3	Unsubmitted cell draft	Escape cancels this cell edit.
Saving change...	Managed operation in progress	Wait for commit or rollback confirmation.
Checking save outcome	Commit acknowledgement unknown	Resolve the outcome before offering retry.
Saved | Calculating	Inputs stored; results pending	Closing does not discard stored inputs.
Saved | Calculations current	Stored and evaluated	Ready to reopen or export.
Save failed	Rollback confirmed; draft retained	Correct the cause, then retry the edit.





1 Persistence proposal

Each submitted edit, paste, or fill is one managed transaction with applicable rules. Save waits for known storage outcomes; stored edits need Undo to reverse.

2 First native save

An untitled workbook needs a native location on its first Save. Temporary-workbook storage, recovery, and lifecycle still need design.

3 Keep versus discard

The close dialog shown is for an unsubmitted draft. In-flight work must resolve or confirm cancellation before closing; it cannot be silently discarded.

4 No blind retries

When commit outcome is unknown, check it before retry. The final persistence model and recovery contract require review before implementation.

14 / Workbook rules and operation outcomes

Explain the affected edit. Distinguish a rejected transaction from committed data with failed follow-up.

DH-US-20

Supplies - DataHorizon _ [] x

File Edit View Insert Format Data Tools Help

New | Open | Save | Undo | Redo | Default font v | 11 v | B | I | Borders | Fill | 0.00 | Sort / Filter

A2:B4 | fx | Notebook

	A	B	C	D	E	F
1	**Item**	**Quantity**	**Unit price**	**Amount**		
2	Notebook	2	3.50	7.00		
3	Pen	4	1.25	5.00		
4	Folder	3	2.00	6.00		

< > | Supplies | Notes | +

Paste rejected Stored data unchanged 100%

Paste rejected - no changes applied x

A required label is blank at A3.

Workbook rule: labels cannot be blank. The paste and its generated edits were rolled back together.

Operation: Supplies!A2:B4

Show details | Return to selection

Saved - follow-up failed x

Your cell changes were saved.

Workbook follow-up failed after commit. Saving again will not undo the committed data or safely repeat the external action.

Follow-up: workbook-defined action

Show details | Return to workbook

1 Required rules apply

Configured workbook handlers run through the service for ordinary GUI operations. Users do not need to author code.

2 Rollback is specific

Affected range, reason, and next action are visible. Technical details expand on request, including the operation identifier.

3 Commit is specific

A failed external follow-up does not turn Saved into Unsaved. Offer recovery only when defined by the action contract.

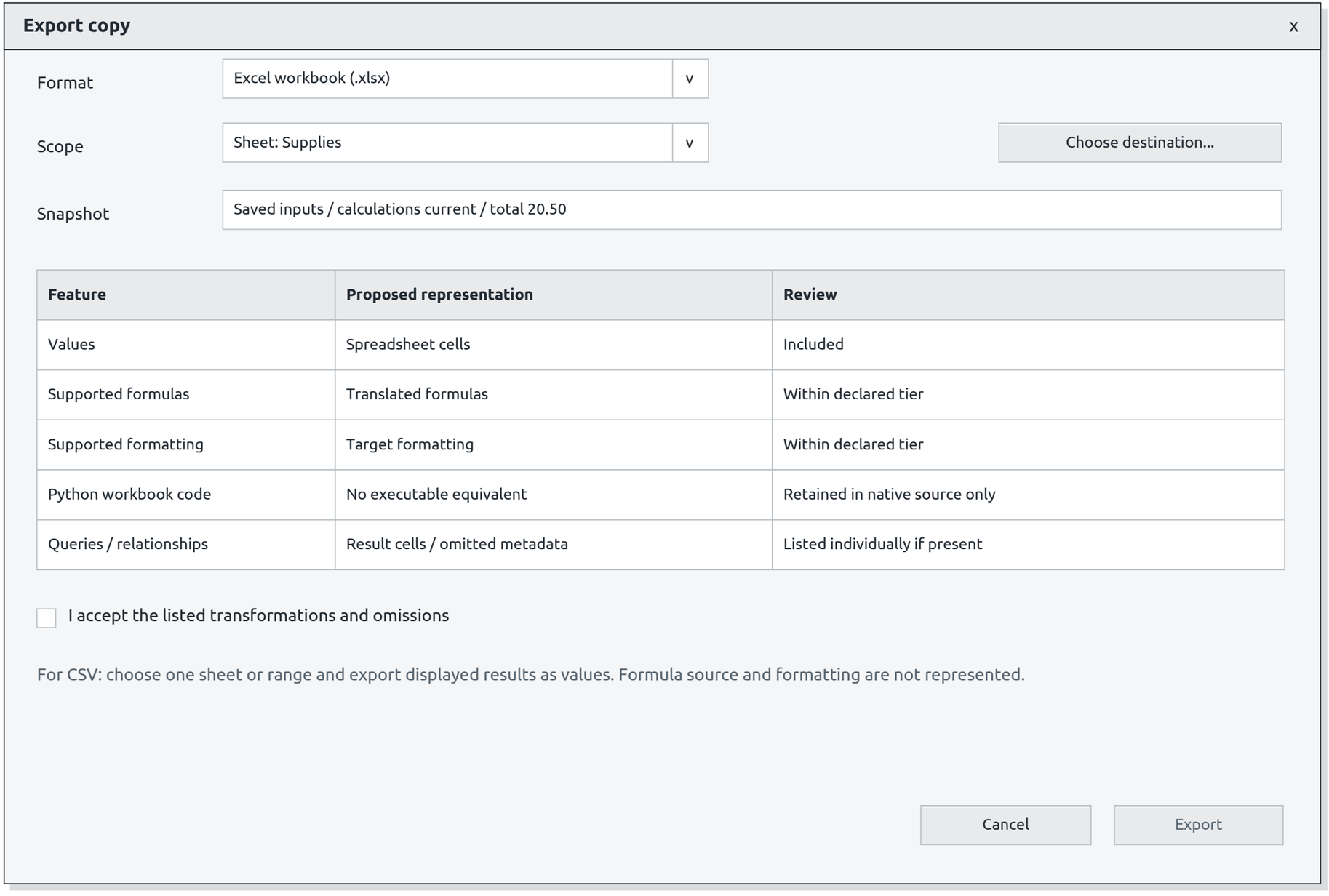
4 External edits differ

On refresh/reopen, reload independent SQLite edits and invalidate derived state. Never replay skipped change handlers; ordinary open events stay separate.

15 / Export a usable copy

Review scope, calculation state, and feature conversion before writing another format.

DH-US-19



1 Consistent snapshot

Export reads a single stored state. Save pending edits first, or explicitly choose the prior saved state.

2 Calculations gate

If results are pending or failed, show affected cells and offer Wait / Recalculate / Cancel. Do not substitute unmarked stale results.

3 Explicit loss review

The final report lists actual features present. Unsupported formulas need a declared fallback; Python code stays in the native workbook.

4 Clear result

Success shows the destination and report. Failure or target size limits show the cause with no silent truncation. Native content is preserved.

16 / Walk the Supplies session

A review script with exact values, visible states, and screen references.

SCREENS

1 Create and enter

03, 04, 08

New workbook > rename Sheet1 to Supplies. Enter the three items, quantities 2 / 4 / 3, and prices 3.50 / 1.25 / 2.00.

2 Calculate and format

05, 06, 07

D2 =B2*C2; fill through D4. D6 =SUM(D2:D4). Display C2:D6 with two decimals. Total = 18.00.

3 Update and inspect

04, 06

Change B3 from 4 to 6. D3 becomes 7.50; D6 becomes 20.50. Dependent cells show Calculating until their result is current.

4 Filter, then restore

10

Quantity > 2 shows Pen and Folder. Notebook remains stored. Clear filters restores all three records.

5 Correct and store

03, 05, 11, 13

Copy a range, paste, undo once, and redo. Save the native workbook; reopen and verify editable formulas and restored formatting.

6 Export and compare

15

Export a supported sheet to Excel and a values snapshot to CSV. Review conversions and confirm the exported total is 20.50.

Check at each step: active cell and source, pending versus stored state, calculation freshness, and recovery path.

17 / Fit to Qt Widgets

Use native desktop building blocks; the spreadsheet behavior still needs application design.

Surface	Qt building blocks	Design responsibility
Window and commands	QMainWindow, QMenuBar, QToolBar, QAction	Same actions in menus, toolbar buttons, and standard shortcuts.
Grid and headers	QTableView, QHeaderView, model + delegate	Custom range selection, fill handle, formula editor, and bounded reads.
Formula and address	QLineEdit in QWidget + layouts	Source editing, validation, and grid focus are coordinated explicitly.
Sheets and side tools	QTabBar, QDockWidget, QSplitter	Overflow, tab movement, resize, hide, and restore panel layout.
Dialogs and fields	QDialog, QFormLayout, QDialogButtonBox	QComboBox, QSpinBox, checkboxes, tables, and platform button roles.
Status and jobs	QStatusBar, QLabel, QProgressBar	Separate commit, calculation, job progress, and operation outcome.

Platform and interaction rules

Use the system palette, fonts, spacing, focus indicators, and style metrics. Light gray chrome here is a neutral schematic, not a prescribed theme. macOS menus may appear in the global menu bar. Button order and file pickers follow the platform.

Standard shortcuts use QKeySequence bindings. Every command is keyboard reachable. Selection and error states use text, borders, or symbols as well as color. At smaller sizes, hide optional docks and overflow toolbar actions while retaining formula entry and status.

Feasibility notes and references

Frozen rows/columns require coordinated views or custom behavior. Qt demonstrates a frozen-column approach. The selected grid design, performance budgets, and precise editing semantics remain open.

Main window / docks	Qt documentation
Table view / model-view	Qt documentation
Frozen-column example	Qt documentation
Dialog button roles	Qt documentation
Platform style / shortcuts	Qt documentation

18 / Coverage and decisions for the next pass

The baseline stories are mapped; several interactions still need product and engine decisions.

Story coverage

DH-US	Goal	Screens
01-02	Create and open	03, 13
03-04	Import and save	12, 13
05-06	Navigate and edit	02, 04
07-08	Copy and fill	05
09-10	Calculate and recover	06
11-12	Rows, columns, sheets	08
13-14	Format and read	07
15-16	Sort and filter	09, 10
17-18	Find and undo	11
19-20	Export and workbook rules	14, 15

Decisions to refine

Save and draft semantics

Automatic storage per managed operation is illustrated. Review this choice, first-save location, temporary workbook recovery, pending drafts, and cancellation.

Grid and reference behavior

Define sort, insertion/deletion, cut, duplicate-sheet references, hidden and filtered paste targets, and supported undo history.

Compatibility and responsiveness

Declare first formula and file-format tiers, value/locale rules, large-file behavior, and measured grid/calculation budgets.

Next wireframe pass

Refine after review: narrow window layout, accessible keyboard paths, empty/loading states, and detailed conversion reports.

Design sources

AGENTS.md; docs/design/README.md; implementation-plan.md; interfaces-and-interoperability.md; data-model.md; architecture.md; python-runtime.md. Story source: output/pdf/datahorizon-everyday-spreadsheet-user-stories.pdf, Draft 0.1, pages 2-12.

Scope: design and documentation only. SQL, Python authoring, remote setup, tables/views mapping, and business-specific workflows are outside this first wireframe set.